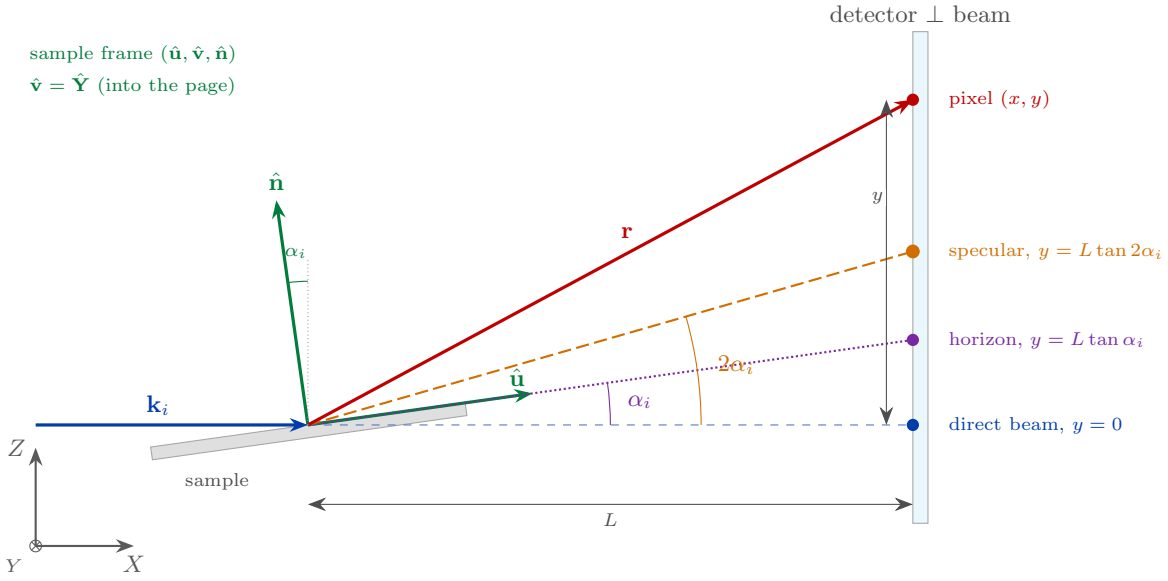
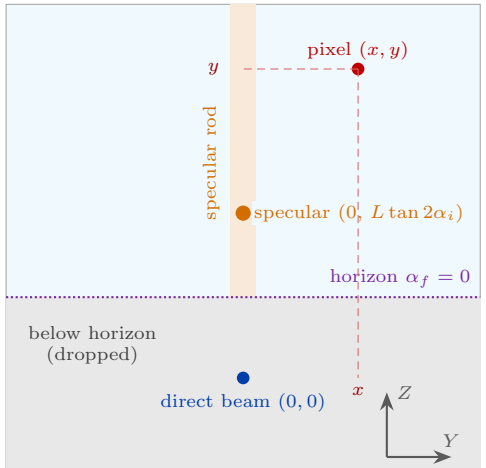
(a) Side view, lab frame



lab frame (X, Y, Z), Y into the page

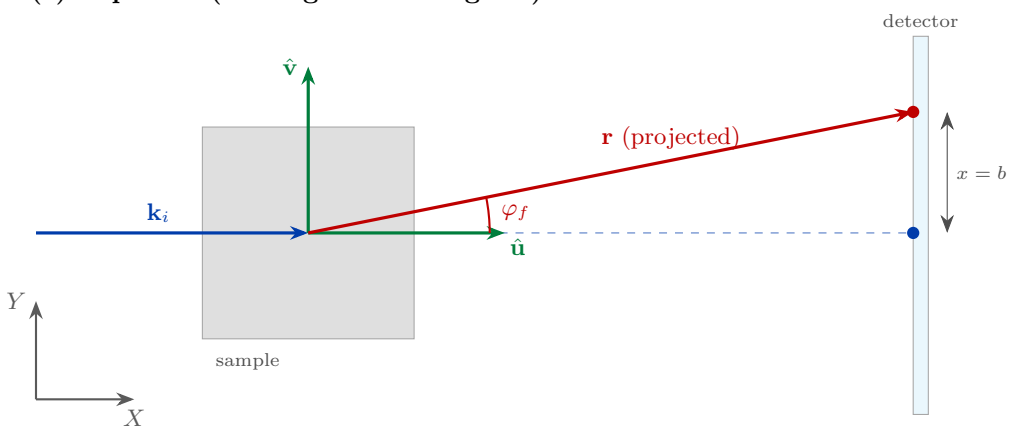
(b) Detector face

seen from behind the detector, looking back at the sample



detector plane = Y–Z plane: x along $Y = \hat{\mathbf{v}}$, y along Z (up).

(c) Top view (looking down along $-Z$)



angles exaggerated ($\alpha_i = 8^\circ$ in the drawing)

(d) Vectors

Sample axes in lab components

$\hat{\mathbf{u}} = (\cos\alpha_i,\ 0,\ \sin\alpha_i), \quad \hat{\mathbf{v}} = (0,\ 1,\ 0), \quad \hat{\mathbf{n}} = (-\sin\alpha_i,\ 0,\ \cos\alpha_i)$

Pixel vector

lab: $\mathbf{r} = L\,\hat{\mathbf{X}} + x\,\hat{\mathbf{Y}} + y\,\hat{\mathbf{Z}}$

sample: $\mathbf{r} = a\,\hat{\mathbf{u}} + b\,\hat{\mathbf{v}} + c\,\hat{\mathbf{n}}$

$a = L\cos\alpha_i + y\sin\alpha_i, \quad b = x, \quad c = y\cos\alpha_i - L\sin\alpha_i$

$R = |\mathbf{r}| = \sqrt{L^2 + x^2 + y^2}$

Exit angles

$\varphi_f = \operatorname{atan2}(b,\ a), \qquad \alpha_f = \operatorname{atan2}\big(c, \sqrt{a^2 + b^2}\big)$

Wave vectors ($k = 2\pi/\lambda$, sample frame)

$\mathbf{k}_i = k\,(\cos\alpha_i,\ 0,\ -\sin\alpha_i)$

$\mathbf{k}_f = \dfrac{k}{R}\,(a,\ b,\ c) = k\,(\cos\alpha_f\cos\varphi_f,\ \cos\alpha_f\sin\varphi_f,\ \sin\alpha_f)$

Scattering vector $\mathbf{q} = \mathbf{k}_f - \mathbf{k}_i$ (components along $\hat{\mathbf{u}}, \hat{\mathbf{v}}, \hat{\mathbf{n}}$)

$q_x = k\Big(\dfrac{a}{R} - \cos\alpha_i\Big), \qquad q_y = k\,\dfrac{b}{R}, \qquad q_z = k\Big(\dfrac{c}{R} + \sin\alpha_i\Big)$